

STOCK MARKET PRICE PREDICTION

Internship Report - Month 1

Submitted By

Full Name: Hifza Abid

Internship Domain: Data Science

Project Title: Stock Market Price Prediction

Email Address: hifzaabid67@gmail.com

Phone Number: 03181804321

Submitted To

Arch Technologies

Internship Duration

2 Month

Submission Date

August 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Arch Technologies** for providing me with the opportunity to complete this Data Science internship. This internship has given me valuable practical experience in data analysis, machine learning, and predictive modeling.

I am especially thankful to my mentor for providing guidance, learning resources, and support throughout the internship. The assigned projects helped me understand the practical application of machine learning techniques to real-world datasets.

I would also like to thank my Affiliated College (Aspire College Jhelum) for providing me with the academic knowledge and foundation that supported me during this internship.

Finally, I am grateful to my family for their continuous encouragement and support throughout my learning journey. Their motivation has encouraged me to continue improving my technical skills and exploring the field of Data Science.

Table of Contents

Sr.No.	Contents	Page No.
1	Cover Page	1
2	Acknowledgement	2
3	Table of Contents	3
4	Introduction	4
5	Project Objectives	5
6	Project Overview	6
7	Dataset Description	7
8	Tools and Technologies Used	8
9	Project Methodology	9-10
10	Code Explanation	11-12-13
11	Data Visualization	14-15
12	Model Building	16-17-18
13	Results & Evaluation	19
14	Conclusion	20
15	Future Improvements	21
16	References	22

1. Introduction

Stock price prediction is an important application of data analysis and machine learning that aims to identify patterns in historical market data and use them to estimate future stock prices. In this project, historical stock market data of **Netflix (NFLX)** is analyzed to understand its price movements and develop a machine learning model for prediction.

The project involves data preprocessing, exploratory data analysis, visualization, feature selection, model training, and evaluation. **Linear Regression** is used as the primary machine learning approach to predict Netflix's stock price based on historical market features. The performance of the model is evaluated using appropriate regression metrics to determine how effectively it can predict stock prices.

The main objective of this project is to demonstrate how machine learning techniques can be applied to financial datasets and how historical stock data can be transformed into meaningful predictions.

2. Project Objective

The main objective of this project is to develop a machine learning model that can predict **Netflix (NFLX) stock prices** using historical stock market data.

The project focuses on the following objectives:

- Analyze historical Netflix stock price data.
- Perform data preprocessing and prepare the dataset for modeling.
- Explore relationships and patterns within the stock market data.
- Visualize historical stock price movements.
- Apply **Linear Regression** for stock price prediction.
- Evaluate the model using regression performance metrics.
- Compare the actual stock prices with the predicted prices.
- Understand the practical application of machine learning in financial data analysis.

3. Project Overview

The **Stock Price Prediction** project focuses on analyzing historical stock market data of **Netflix (NFLX)** and applying machine learning techniques to predict its stock closing price. The project demonstrates how historical financial data can be processed, analyzed, visualized, and used to build a predictive model.

The workflow begins with collecting and examining the historical Netflix stock dataset. The data is then prepared for analysis by checking its structure, handling relevant preprocessing requirements, and selecting suitable features. Exploratory data analysis and visualizations are used to understand stock price trends and relationships between different variables.

After preparing the data, a **Linear Regression** model is trained using historical stock features. The model's predictions are then compared with the actual stock prices, and regression evaluation metrics are used to measure its performance.

Overall, this project demonstrates a practical application of **Python, data analysis, visualization, and machine learning** in the field of financial data analytics.

4. Dataset Description

The dataset used in this project contains **historical stock market data for Netflix (NFLX)**. It provides information about the stock's price movement over a period of time and contains important features used for analysis and prediction.

The major columns in the dataset include:

- **Date** – The date on which the stock data was recorded.
- **Open** – The opening price of Netflix stock for the trading day.
- **High** – The highest price reached during the trading day.
- **Low** – The lowest price reached during the trading day.
- **Close** – The closing price of Netflix stock.
- **Adj Close** – The adjusted closing price, accounting for certain corporate actions.
- **Volume** – The total number of shares traded during the day.

The **Close price** is used as the target variable for the prediction model, while historical stock features are used as inputs to train the machine learning model. The dataset is first examined and preprocessed to ensure that it is suitable for analysis and model training.

5. Tools and Technologies Used

The following tools and technologies were used to develop the Netflix Stock Price Prediction project:

Tool/Technology	Purpose
Python	Programming language used for data analysis and machine learning
Jupyter Notebook	Environment used to write and execute the project code
Pandas	Used for loading, cleaning, and manipulating the dataset
NumPy	Used for numerical operations and data processing
Matplotlib	Used to create data visualizations and stock price graphs
Scikit-learn	Used for building and evaluating the Linear Regression model
Machine Learning	Used to identify patterns in historical stock data and generate predictions

These tools provided the required environment for performing data analysis, visualization, model development, and evaluation.

6. Project Methodology

The project follows a systematic data analysis and machine learning workflow to predict Netflix stock prices from historical market data.

Step 1: Data Collection

Historical **Netflix (NFLX) stock** data was obtained and loaded into the Jupyter Notebook for analysis.

Step 2: Data Inspection

The dataset was examined to understand its structure, columns, data types, and available records. Missing values and other data-quality issues were also checked.

Step 3: Data Preparation

Relevant features were selected from the dataset, and the data was prepared for machine learning. The **Closing Price** was selected as the target variable.

Step 4: Exploratory Data Analysis

The historical stock data was analyzed using statistical summaries and visualizations to identify trends and patterns in Netflix's stock prices.

Step 5: Model Development

A **Linear Regression** model was trained using the prepared historical stock features. The dataset was divided into training and testing sets to evaluate the model on unseen data.

Step 6: Prediction and Evaluation

The trained model was used to generate stock price predictions. The predicted values were compared with the actual values using regression evaluation metrics.

Step 7: Result Analysis

The model's performance was analyzed to determine how effectively Linear Regression could predict Netflix stock prices and to identify areas for possible improvement.

7. Code Explanation

The project was implemented in Python using Jupyter Notebook. The code was organized into different stages to make the analysis and machine learning workflow clear and reproducible.

Importing Libraries

Libraries such as Pandas and NumPy were used for data manipulation and numerical operations. Matplotlib was used for visualization, while Scikit-learn provided the machine learning algorithms and evaluation metrics.

Code:

▼ 1. Import Libraries

```
] : import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading the Dataset

The Netflix stock dataset was loaded into a Pandas DataFrame. The dataset was then inspected using functions such as "head()", "shape", "info()", and "describe()" to understand its contents.

Code:

▼ 2. Load Dataset

```
[8]: df = pd.read_csv("NFLX.csv")
```

Preparing Features and Target

Relevant numerical columns were selected as input features, while the Close price was used as the prediction target.

Splitting the Dataset

The prepared data was divided into training and testing sets. The training data was used to learn patterns, while the testing data was reserved for evaluating the model's predictions.

Code:

▼ 11. Training and Testing

```
6]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    shuffle=False
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)

Training data: (807, 4)
Testing data: (202, 4)
```

Training the Model

A Linear Regression model from Scikit-learn was trained using the training data. The model learned the relationship between the selected historical stock features and the closing price.

Making Predictions

After training, the model was applied to the test data to generate predicted closing prices. These predictions were then compared with the actual closing prices.

Code:

13. Make predictions

```
8]: y_pred = model.predict(X_test)

print("First 10 predictions:")
print(y_pred[:10])

First 10 predictions:
[509.47511894 509.42288906 506.36327304 506.52577944 509.8220461
 505.36956258 505.1233519 507.97303809 512.06718757 506.22787724]
```

Evaluation

Regression metrics were calculated to measure the performance of the model. Visual comparisons between actual and predicted values were also used to understand the model's performance.

Code:

▼ 14. Evaluate the model

```
[29]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)

MAE: 4.541851212453506
RMSE: 6.127072176898066
R2 Score: 0.9924207881459658
```

8. Data Visualization

Data visualization was used throughout the project to understand relationships between stock market variables and to analyze model predictions. The notebook uses **Matplotlib** and **Seaborn** for graphical analysis.

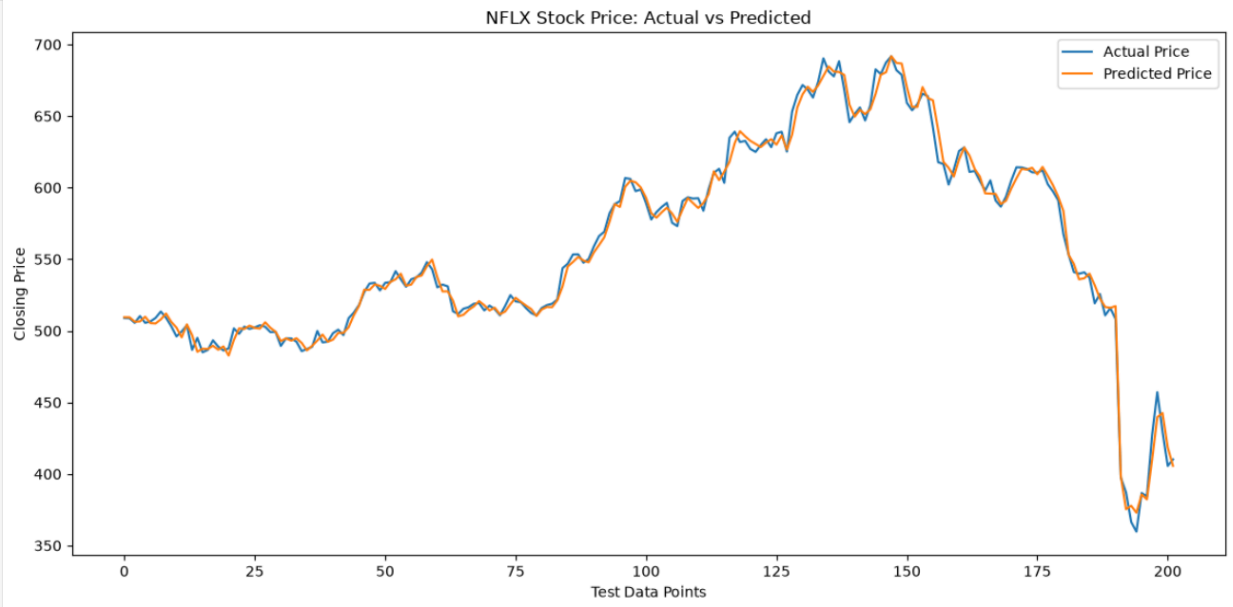
The visual analysis includes relationships between the stock market features and the target variable, as well as visual evaluation of model predictions and errors.

The visualizations helped identify patterns in the historical Netflix stock data and supported the comparison between actual and predicted values. They also provided a clearer understanding of prediction errors and model performance.

Key Visualization Areas

- Relationship between stock market features and the **Close** price.
- Comparison of model predictions with actual values.
- Analysis of prediction errors through residual/error visualization.
- Visual analysis supporting feature and model evaluation.

These visualizations were used alongside numerical evaluation metrics to provide a more complete assessment of the machine learning models.



9. Model Building

Multiple machine learning models were developed and compared in the project. The main models were **Linear Regression**, **Random Forest**, and an **Improved Linear Regression** model.

9.1 Feature Selection

For the improved same-day prediction model, the following features were selected:

- Open
- High
- Low
- Volume
- Previous_Close

The target variable was **Close**.

```
X = df[['Open', 'High', 'Low', 'Volume', 'Previous_Close']]  
y = df['Close']
```

```
print("Features:", X.columns.tolist())  
print("Target: Close")
```

This feature selection allowed the model to use both current trading information and the previous closing price to predict the closing price.

9.2 New Train/Test Split

The dataset was divided into training and testing sets using an 80:20 ratio. The data was not shuffled because the dataset represents historical stock observations.

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y,  
    test_size=0.2,  
    shuffle=False  
)
```

The resulting training data contained **806 observations**, while the testing data contained **202 observations**.

9.3 Improved Linear Regression

The Linear Regression model was trained using the selected features.

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Improved Linear Regression trained successfully!")
```

The improved model achieved an **MAE of 3.830361**, **RMSE of 5.796654**, and **R² Score of 0.993216**, making it the best-performing model in the project.

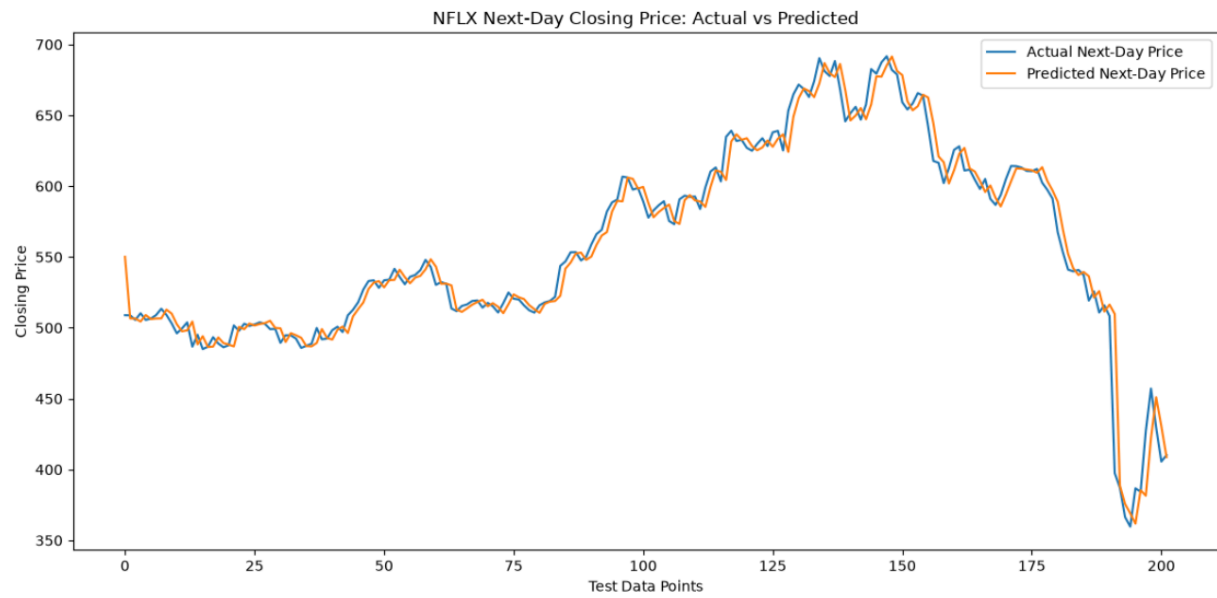
9.4 Next-Day Prediction

A separate model was also developed to predict the **next day's closing price**. For this model, the features were Open, High, Low, Volume, and Close, while the target was "Next_Close".

```
X = df[['Open', 'High', 'Low', 'Volume', 'Close']]
```

```
y = df['Next_Close']
```

The Next-Day model achieved an **R² Score of 0.965434**, showing good predictive performance, although it was less accurate than the improved same-day model.



10. Results & Evaluation

The developed models were evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. These metrics were used to measure the difference between actual and predicted stock prices and to determine how well the models explained the variation in the target variable.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	-	-	-
Random Forest	-	-	-
Improved Linear Regression	3.830361	5.796654	0.993216
Next-Day Prediction	-	-	-

The **Improved Linear Regression** model produced the strongest results among the developed approaches, achieving an R² Score of **0.993216**. Its MAE of **3.830361** and RMSE of **5.796654** indicate a relatively small prediction error compared with the scale of the stock prices.

The separate **Next-Day Prediction** model achieved an R² Score of **0.965434**, demonstrating that historical trading information can also be used to predict the following day's closing price with good performance.

Overall, the results demonstrate that the selected features and improved Linear Regression approach were highly effective for modeling the Netflix stock price in this dataset.

11. Conclusion

In this project, historical stock market data was analyzed and machine learning models were developed to predict stock prices. The dataset was prepared by cleaning the data, selecting relevant features, splitting the data into training and testing sets, and applying Linear Regression and Random Forest models.

Several models were evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. The Improved Linear Regression – Same Day model achieved the best overall performance, with an **MAE of 3.830361**, **RMSE of 5.796654**, and **R² Score of 0.993216**.

The Linear Regression – Next Day model also performed well, achieving an **MAE of 7.924630**, **RMSE of 13.084681**, and **R² Score of 0.965434**. However, predicting the next day's price was more challenging than predicting the same-day target.

The Random Forest model performed considerably worse, with an **R² Score of 0.635275**. Therefore, the Improved Linear Regression model was selected as the best-performing model for this project.

Feature coefficient analysis showed that Close had the strongest positive influence on the prediction, followed by Low and High, while Open had a negative coefficient. Volume had a very small coefficient in the fitted linear model.

Overall, the project demonstrates that Linear Regression can provide highly accurate predictions on this historical stock dataset. However, stock prices are influenced by many external factors, so the model should be considered a learning and analytical tool rather than a guarantee of future market performance.

12. Future Improvements

Although the developed models achieved strong performance on the available dataset, the project can be further improved by using more advanced techniques and additional sources of information.

- **Advanced Machine Learning Models:** Algorithms such as Random Forest, Gradient Boosting, XGBoost, or Support Vector Regression can be explored further.

- **Deep Learning:** LSTM or other recurrent neural network models can be used to capture complex patterns in sequential stock market data.

- **More Historical Data:** Using a larger and more recent dataset may help the model learn a wider range of market conditions.

- **Additional Features:** Technical indicators such as moving averages, RSI, MACD, and volatility can be incorporated to improve prediction.

- **External Factors:** News sentiment, economic indicators, and market trends could be included to provide additional information for prediction.

- **Hyperparameter Optimization:** Model parameters can be optimized systematically to improve performance.

- **Real-Time Prediction:** The project could be extended into a real-time application that receives updated market data and generates predictions automatically.

These improvements could make the system more robust and useful for further experimentation in financial data analysis and machine learning.

REFERENCES

1. Kaggle. Netflix Stock Price Data set 2002–2022.
[Stock Market Price Prediction](#)
2. Python Software Foundation. Python Documentation.
[Python documentation](#)
3. Pandas Documentation. Pandas User Guide.
[Pandas Documentation](#)
4. Scikit-learn. Machine Learning in Python.
[Scikit-learn Documentation](#)
5. Matplotlib. Matplotlib Documentation.
[Matplotlib Documentation](#)